Austin Huang

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SUMMARY

Robotics Systems Engineer focused on production-grade automation across robotics, vision, and software integration. Brings experience in system architecture, reliability engineering, and cross-functional deployment under real-world constraints. Recently promoted into a leadership role following the company's first U.S. commercial deployment.

SKILLS

Core Strengths: Hardware–Software Integration; Reliability Engineering; System Architecture; High-Throughput Workflow Design; Data-Driven Optimization; Cross-Functional Execution; Customer-Facing Deployment; Technical Documentation.

Technical Skills: Robotics Integration (ABB Robots, AGV/AMR Systems); ROS/ROS2; Vision Systems (OpenCV, YOLO); Motion Planning; API Integration; Python; C++; MATLAB; Mechanical Design (Creo); System Calibration; Linux (Ubuntu); Git.

EXPERIENCE

XtalPi Inc., Boston, MA
Robotics Systems Engineer *(Promoted to Manager / Lead, Automation Systems – North America)* *August 2024 – Present*

- Owned end-to-end deployment and stabilization of a production-grade automated chemistry R&D platform, defining system architecture, integration logic, and deployment readiness for U.S. commercial rollout.
- Served as the primary on-site system owner at a client, recovering system performance under real-world constraints and improving task evaluation pass rates from 59% to 100% and target hit rates from 29% to 98%.
- Developed a monocular contour-based calibration method for end-effector pose correction, improving alignment accuracy while reducing hardware dependency and deployment complexity for solid-dispensing workflows.
- Led system-level reliability improvements across task coordination, vision-guided manipulation, and failure handling, reducing operational failures by 94% and increasing readiness for scalable production deployment.
- Defined system interfaces across robotics, control logic, and API layers, enabling reliable cross-system integration and consistent production behavior.
- Supported the company's first U.S. commercial deployment through live system validation, technical problem-solving, and system stabilization, contributing to customer adoption and subsequent multi-million-dollar expansion.

Axle Informatics, Rockville, MD
Automation Research Associate, *Internship* *September 2023 – December 2023*

- Owned the design and deployment of a High-Density Storage (HDS) subsystem, enabling reliable integration with Omron mobile robots and PF400 robot arms and contributing to a 37% reduction in lab process time.
- Improved the HDS perception reliability from 53% to 98% by strategic modifications to image preprocessing filters and implementing advanced shape detection algorithms using OpenCV and Python.

Laboratory for Computational Sensing and Robotics, Baltimore, MD
Research Assistant, *Advisor: Russell H. Taylor* *May 2023 – September 2023*

- Optimized malaria vaccine production system setup time by 62% using deep learning-based computer vision.
- Improved ROI detection accuracy to 96% using YOLOv5 and PyTorch-based class activation mapping, achieving 95% vaccine quality prediction accuracy.

Dynacolor, Taipei, Taiwan
R&D Engineer *May 2019 – March 2021*

- Engineered mechanical components for a 6-DOF surgical camera system meeting IP66/67 and IK10 requirements.
- Reduced defects by over 80% through SOP development, root-cause analysis, and process optimization.

PROJECTS

Sensor-based Robot Arm Manipulation, Johns Hopkins University *September 2022 - December 2022*

- Improved UR5 manipulation performance by reducing control time by over 50% and increasing RRT planning efficiency by 73% through control tuning in MATLAB and ROS and k-d tree implementation in C++.
- Implemented hand-eye calibration using the Park-and-Martin method, achieving 0.2 mm end-effector error.

EDUCATION

Johns Hopkins University, Baltimore, MD
Master of Science in Engineering in Robotics (GPA: 3.9) *August 2022 - May 2024*

National Taiwan University, Taipei, Taiwan
Bachelor of Science in Mechanical Engineering *September 2014 - June 2018*